A puzzle about exclusives

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What's in the shed?

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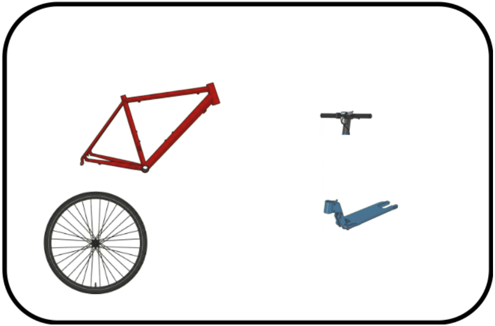
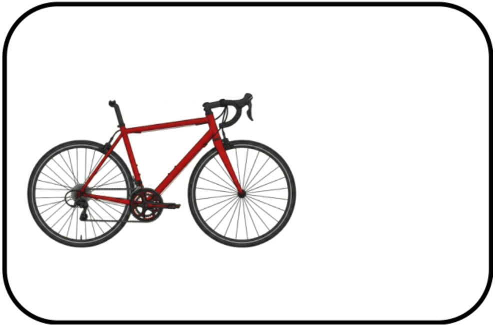


What's in the shed?



What's in the shed?

(1) Only the bike is in the shed. [Assumption]

(2) So the bike is in the shed. [From 1]

(3) If the bike is in the shed, then at least one of its parts is in the shed. [Assumption]

(4) So at least one of the bike's parts is in the shed. [From (2), (3)]

(5) The bike's parts are numerically distinct from the bike. [By Lebiniz's Law]

(6) So something distinct from the bike is in the shed; i.e., at least one of its parts. [From 4, 5]

(7) So not only the bike is in the shed. [From (6)]

Plan

Against the inference from (6) to (7)

Domain shift

A scalar analysis

Against the inference from (6) to (7)

If the inference is valid, then (6) should be inconsistent with (1).

To show: (6) is consistent with (1).

Against the inference from (6) to (7)

Felicitous rejection: If S and S' are inconsistent, then it is felicitous to reject S with S'.

Against the inference from (6) to (7)

Examples:

(8) X: Grass is blue.
Y: # No, grass is short.

Against the inference from (6) to (7)

Felicitous Rejection shows that (6) and (1) are consistent (inspired by Kratzer2012).

(1) Only the bike is in the shed.

(9) # No, the bike's seat is also in the shed.

(10) # No, the bike's handle is also in the shed.

(11) # No, the bike's parts are also in the shed.

Against the inference from (6) to (7)

All of (9)–(11) entail that something distinct from the bike is in the shed (via Leibniz's Law).

So if (6) were inconsistent with (1), then (9)–(11) should be felicitous (by Felicitous Rejection).

Against the inference from (6) to (7)

Domain shift

Domain shift:

(6) and (1) are interpreted relative to different domains, and this makes the two consistent. The inference from (6) to (7) is like a fallacy of equivocation. (Casati and Varzi1999; Fine2006; Miller2021; Heller p.c.)

Domain shift

A toy semantic value for (1), based on Horn (1969):

$[\![\text{Only B is in the shed}]\!]^D$ presupposes that B is in the shed.

$[\![\text{Only B is in the shed}]\!]^D = 1$ iff $\forall x \in D(x \neq \text{B} \rightarrow$
$\neg x$ is in the shed)

Domain shift

$$D_1 = \{\mathsf{B}, \mathsf{B}_1, \ldots, \mathsf{B}_\mathsf{n}, \mathsf{S}, \mathsf{S}_1, \ldots, \mathsf{S}_\mathsf{n}\}$$

$$D_2 = \{\mathsf{B}, \mathsf{S}\}$$

Domain shift

$[\![\text{Something distinct from the bike is in the shed}]\!]^{D_1}$ is true iff at least one of $\mathsf{B}_1, \ldots, \mathsf{B}_\mathsf{n}, \mathsf{S}, \mathsf{S}_1, \ldots, \mathsf{S}_\mathsf{n}$ is in the shed.

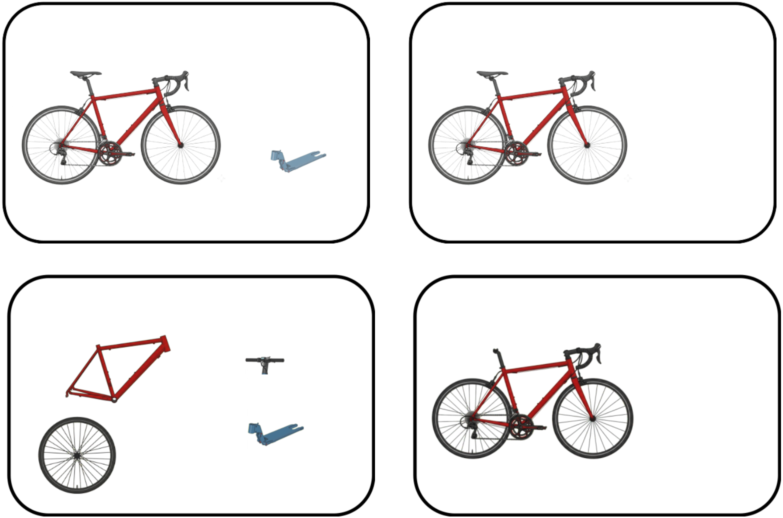
$[\![\text{Only B is in the shed}]\!]^{D_2}$ is true iff S is not in the shed.

The two are consistent.

Domain shift

To show: $[\![\text{Only B is in the shed}]\!]^{D_2}$ is not the preferred interpretation in this context (it might not even be available).

Domain shift



Domain shift

What's in the shed?

(12) Only the bike is in the shed, so the scooter's handle/deck/back wheel is not in the shed.

(13) If only the bike is in the shed, then the scooter's handle/deck/back wheel is not in the shed.

Against the domain-shift analysis

If (1) is interpreted relative to D_2, then (12) and (13) should express invalid inferences. Compare with:

(14) # The scooter is not in the shed, so the scooter's handle/deck/back wheel is not in the shed.

(15) # If the scooter is not in the shed, then the scooter's handle/deck/back wheel is not in the shed.

Domain shift

Domain shift analysis for (7):

$[\![\text{Not only B is in the shed}]\!]^{D_2}$ presupposes that B is in the shed.

$[\![\text{Not only B is in the shed}]\!]^{D_2}$ asserts that S is in the shed.

Against the domain-shift analysis

If (7) is interpreted relative to D_2, then (16) should express a valid inference.

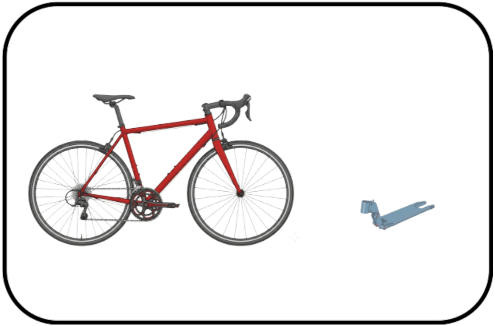
(16) # Not only the bike is in the shed, so everything is in the shed.

But this inference is invalid in the present context. Evidence for this is that the following rejection of (16) is reasonable:

(17) That doesn't follow; $\mathsf{S}_1/\ldots/\mathsf{S}_\mathsf{n}$ might not be in the shed.

Domain shift

What's in the shed?

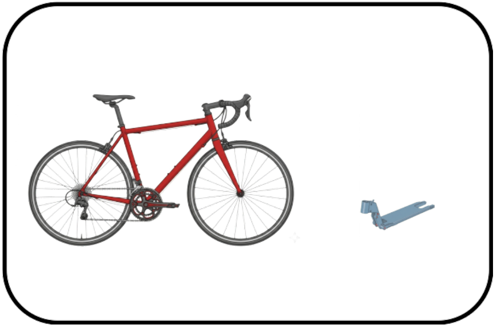


Domain shift

What's in the shed?

(1) ✗ Only the bike is in the shed

(7) ✓ Not only the bike is in the shed



Domain shift

Conclusion: $[\![\text{Only B is in the shed}]\!]^{D_2}$ is not the preferred interpretation in this context.

A scalar analysis

C = a state of the conversation

I_C = the conversation's common ground

QUD_C = the question under discussion in the conversation

$\geq_C$ = a contextually determined strength ranking on QUD_C

A scalar analysis

Coppock and Beaver's (2014) entry for "only":

(18) $\text{min}_C(p) = \lambda w.\exists p' \in \text{QUD}_C.p'(w) \wedge p' \geq_C p$

(19) $\text{max}_C(p) = \lambda w.\forall p' \in \text{QUD}_C.p'(w) \rightarrow p \geq_C p'$

(20) $[\![\text{only}]\!]^C = \lambda p.\lambda w : \text{min}_C(p)(w).\,\text{max}_C(p)(w)$

A scalar analysis

$\min_C(p)$ says that some answer to QUD at least as strong as p is true.

$\max_C(p)$ says that no answer to QUD stronger than or incomparable to p is true.

$\ulcorner$only $\phi\urcorner$ presupposes that an answer to QUD at least as strong as $[\![\phi]\!]^C$ is true, and asserts that nothing stronger than $[\![\phi]\!]^C$ or incomparable with it is true.

A scalar analysis

(21) Cora is only a major.

$$\begin{array}{c}\text{general}\\ \downarrow\\ \text{major}\\ \downarrow\\ \text{lieutenant}\end{array}$$

A scalar analysis

b = B is in the shed
b_1 = B_1 is in the shed
$\vdots$

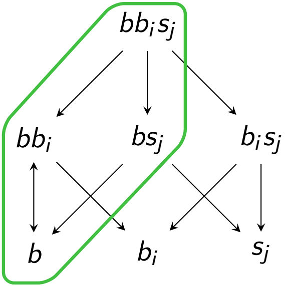
A scalar analysis

$\mathrm{QUD}_{\mathrm{S}} =$ all combinations of $b, b_1, \ldots, b_n, s, s_1, \ldots, s_n$

A scalar analysis

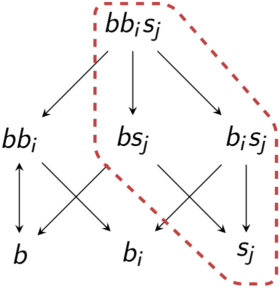
$$
\begin{tikzcd}
 & bb_is_j \arrow[dl] \arrow[d] \arrow[dr] & \\
bb_i \arrow[d, leftrightarrow] \arrow[dr] & bs_j \arrow[dl] \arrow[dr] & b_is_j \arrow[dl] \arrow[d] \\
b & b_i & s_j
\end{tikzcd}
$$

A scalar analysis



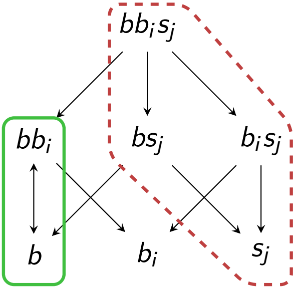
$[\![$Only the bike is in the shed$]\!]^C$ presupposes that at least b is true.

A scalar analysis



⟦Only the bike is in the shed⟧^{C} asserts that at most b is true.

A scalar analysis



Presupposition + Assertion

A scalar analysis

(1) is consistent with "$B_1/\ldots/B_n$ is in the shed"

(1) is consistent with (6)—"Something distinct from the bike is in the shed".

(1) is inconsistent with "$S/S_1/\ldots/S_n$ is in the shed"

A scalar analysis

Is there a natural relation between propositions corresponding to this choice of $\geq_S$?

$$\begin{tikzcd}
 & bb_i s_j \arrow[dl] \arrow[d] \arrow[dr] & \\
bb_i \arrow[d, leftrightarrow] \arrow[dr] & bs_j \arrow[dl] \arrow[dr] & b_i s_j \arrow[dl] \arrow[d] \\
b & b_i & s_j
\end{tikzcd}$$

A scalar analysis

First stab: $\geq_S$ is local entailment (Yablo2006).

A scalar analysis

P **locally entails** Q in *w* iff Q is true in every situation in *w* in which P is true.

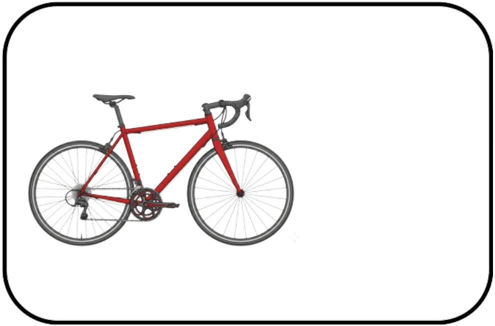
A scalar analysis

Roughly, situations are parts of worlds, and worlds are maximal consistent situations.

A scalar analysis

Assumption: if x is an attached part of y, and y is fully in a certain region of space at w, then every situation in w in which y is fully in that region is a situation in which x is fully in that region.

A scalar analysis



A scalar analysis

Every situation in w_0 in which B is in the shed is a situation in which its parts are in the shed. None of S or its parts are in the shed.
Then:

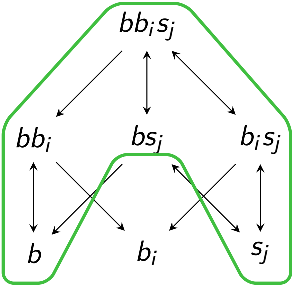
> At w_0: b locally entails $b_1, \ldots, b_n$ and their conjunctions.
>
> At w_0: $s, s_1, \ldots, s_n$ and anything that has them as conjuncts locally entail everything.

A scalar analysis

Where $\geq_S$ is local entailment at w_0:

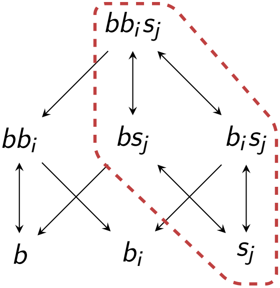
$$\begin{tikzcd}
 & bb_i s_j \arrow[dl] \arrow[d, leftrightarrow] \arrow[dr, leftrightarrow] & \\
bb_i \arrow[d, leftrightarrow] \arrow[dr] & bs_j \arrow[dl] \arrow[dr, leftrightarrow] & b_i s_j \arrow[dl] \arrow[d, leftrightarrow] \\
b & b_i & s_j
\end{tikzcd}$$

A scalar analysis



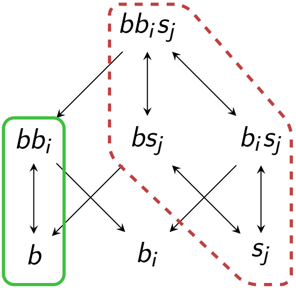
$[\![$Only the bike is in the shed$]\!]^C$ presupposes that at least b is true.

A scalar analysis



$[\![$Only the bike is in the shed$]\!]^C$ asserts that at most b is true.

A scalar analysis

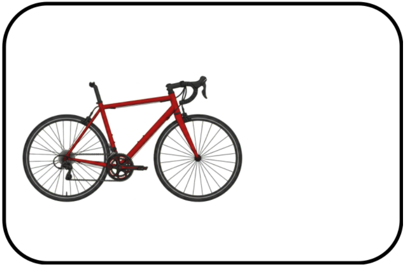


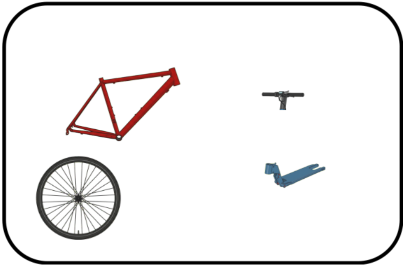
Presupposition + Assertion

A scalar analysis

What's so special about w_0 or relevantly similar worlds?

What's in the shed?





A scalar analysis

Amendment: $\geq_C$ can be world-sensitive, so min and max should be world-sensitive too.

A scalar analysis

(22) $\min^*_C(p) = \lambda w.\exists p' \in \mathsf{QUD}_C[p'(w) \wedge (p' \geq_{C,w} p)]$

(23) $\max^*_C(p) = \lambda w.\forall p' \in \mathsf{QUD}_C[p'(w) \rightarrow (p \geq_{C,w} p')]$

$\quad = \lambda w.\forall p' \in \mathsf{QUD}_C[(p \not\geq_{C,w} p') \rightarrow \neg p'(w)]$

(24) $[\![\text{only}]\!]^C = \lambda p.\lambda w : \min^*_C(p)(w)\,.\ \max^*_C(p)(w)$

A scalar analysis

$\text{min}^*_C(p)$ is true at w iff some answer to QUD at least as strong as p at w is true at w.

$\text{max}^*_C(p)$ is true at w iff no answer to QUD stronger than or incomparable to p at w is true at w.

$\ulcorner$only $\phi\urcorner$ presupposes $\text{min}^*_C([\![\phi]\!]^C)$ and asserts $\text{max}^*_C([\![\phi]\!]^C)$.

A scalar analysis

(21) Cora is only a major.

$$\begin{array}{c} \text{general} \\ \downarrow \\ \text{captain} \\ \downarrow \\ \text{major} \end{array} \qquad\qquad \begin{array}{c} \text{general} \\ \downarrow \\ \text{major} \\ \downarrow \\ \text{captain} \end{array}$$

A scalar analysis

$$\boxed{\begin{array}{c}\text{general}\\ \downarrow \\ \text{captain}\\ \downarrow \\ \text{major}\end{array}} \qquad\qquad \begin{array}{c}\boxed{\begin{array}{c}\text{general}\\ \downarrow \\ \text{major}\end{array}}\\ \downarrow \\ \text{captain}\end{array}$$

$[\![$Cora is only a major$]\!]^C$ presupposes that, whatever the ordering of ranks is, Cora's rank is at least as high as major.

A scalar analysis

general
$\downarrow$
captain
$\downarrow$
major

general
$\downarrow$
major
$\downarrow$
captain

$\llbracket$Cora is only a major$\rrbracket^{C}$ asserts that, whatever the ordering of ranks is, Cora's rank is at most as high as major.

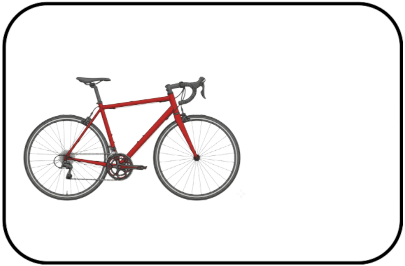
A scalar analysis

$$\begin{array}{ccc} \text{general} & \qquad & \text{general} \\ \downarrow & & \downarrow \\ \text{captain} & & \boxed{\text{major}} \\ \downarrow & & \downarrow \\ \boxed{\text{major}} & & \text{captain} \end{array}$$

Presupposition + Assertion

A scalar analysis

$b \geq_{C,w} b_1, \ldots, b_n$ at worlds at which $\mathrm{B}_1, \ldots, \mathrm{B}_n$ are attached to the bike.



$$\begin{tikzcd}
 & bb_i s_j \arrow[dl] \arrow[d, leftrightarrow] \arrow[dr, leftrightarrow] & \\
bb_i \arrow[d, leftrightarrow] \arrow[dr] & bs_j \arrow[dl] \arrow[dr, leftrightarrow] & b_i s_j \arrow[dl] \arrow[d, leftrightarrow] \\
b & b_i & s_j
\end{tikzcd}$$

A scalar analysis

$b \not\geq_{s,w} a_i$ in worlds in which B is in the shed, but B_i is detached from B.

$$\begin{tikzcd}
& & bb_is_j \arrow[dll, <->] \arrow[d] \arrow[drr, <->] & & \\
bb_i \arrow[d] \arrow[drr, <->] & & bs_j \arrow[dll] \arrow[drr] & & b_is_j \arrow[dll, <->] \arrow[d] \\
b & & b_i & & s_j
\end{tikzcd}$$

A scalar analysis

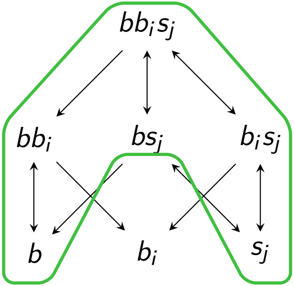
$$\begin{tikzcd}
 & bb_i s_j \arrow[dl] \arrow[d, leftrightarrow] \arrow[dr, leftrightarrow] & \\
bb_i \arrow[d, leftrightarrow] \arrow[dr] & bs_j \arrow[dl] \arrow[dr, leftrightarrow] & b_i s_j \arrow[dl] \arrow[d, leftrightarrow] \\
b & b_i & s_j
\end{tikzcd}$$

b locally entails b_i

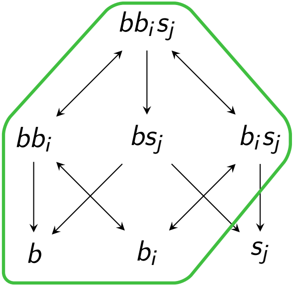
$$\begin{tikzcd}
 & bb_i s_j \arrow[dl, leftrightarrow] \arrow[d] \arrow[dr, leftrightarrow] & \\
bb_i \arrow[d] \arrow[dr, leftrightarrow] & bs_j \arrow[dl] \arrow[dr] & b_i s_j \arrow[dl, leftrightarrow] \arrow[d] \\
b & b_i & s_j
\end{tikzcd}$$

b does *not* locally entail b_i

A scalar analysis



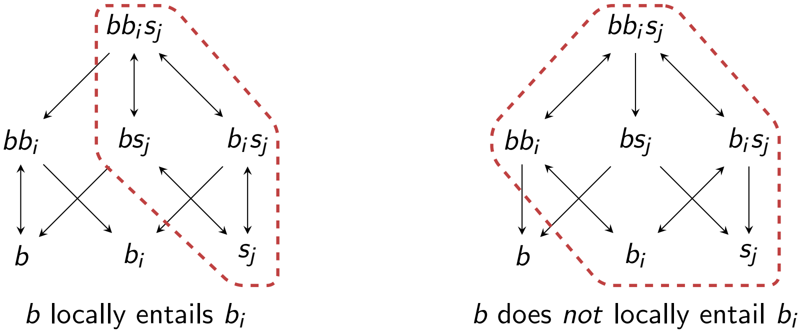
b locally entails b_i



b does *not* locally entail b_i

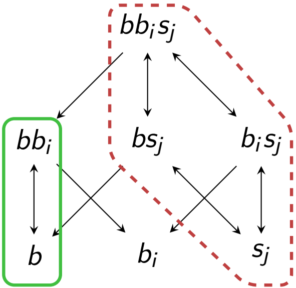
$[\![$ Only the bike is in the shed$]\!]^C$ presupposes that something in QUD_C which locally entails b is true (at least b is true).

A scalar analysis

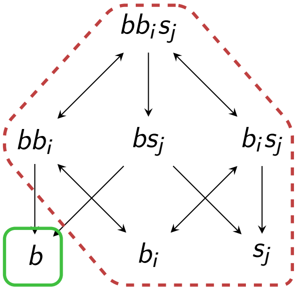


$[\![$ Only the bike is in the shed$]\!]^C$ asserts that no answer to QUD_C which is not locally entailed by b is true (at most b is true).

A scalar analysis



b locally entails b_i



b does *not* locally entail b_i

Presupposition + Assertion

A scalar analysis

$[\![$Only B is in the shed$]\!]$ is compatible with part B_i of the bike being in the shed, but excludes part S_j of the scooter being in the shed.

Conclusion

Diagnosed the problem with the initial argument.

Argued that the domain-shift view is not successful.

Presented a scalar analysis using the notion of local entailment.

Conclusion

Bad news for some restrictions on quantificational domains

Casati and Varzi on inventories (1999, p. 112):

> *[E]very admissible way of drawing up an inventory must satisfy a non-redundancy condition: If x properly overlaps y and y is included in the inventory, then x is not itself to be included. This avoids double counting.*

References I

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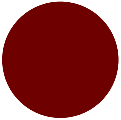
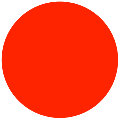
Appendix A

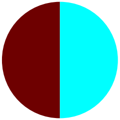
(25) The ball is only red.



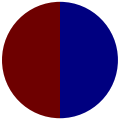
(a) Oxblood



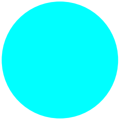
(b) Scarlet



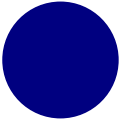
(c) Oxblood and cyan



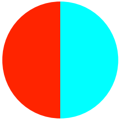
(d) Oxblood and navy



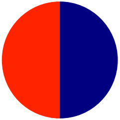
(e) Cyan



(f) Navy



(g) Scarlet and cyan



(h) Scarlet and navy

Appendix B

Following Fine (2017), let a modalized state space be a triple $\langle S, S^{\Diamond}, \sqsubseteq \rangle$ such that:

$S \neq \varnothing$ and each subset of S has a least upper bound in S.

$S^{\Diamond} \subseteq S$ such that if $s \in S^{\Diamond}$ and $t \sqsubseteq s$, then $t \in S^{\Diamond}$.

$\sqsubseteq$ is a partial order on S.

Appendix B

Intuitively,

S is a set of situations.

$S^\Diamond$ is the set of *possible*/*consistent* situations.

$\sqsubseteq$ is a parthood relation on S.

Appendix B

Where $X \subseteq S$ with members $x_1, x_2, \ldots$, we let $x_1 \sqcup x_2 \sqcup \ldots$ be their least upper bound, i.e., their **fusion**.

Appendix B

Possible worlds are maximal consistent fusions of situations.

I.e., $w \in S$ is a possible world iff $w \in S^{\Diamond}$ and any state in $S^{\Diamond}$ is either part of w or incompatible with w.

Appendix B

Propositions are ordered pairs of sets of situations.

Appendix B

$$P = \langle P^{+}, P^{-} \rangle$$

P^{+} is the set of P's verifiers/truthmakers.

P^{-} is the set of P's falsifiers/falsemakers.

Appendix B

$[\![\cdot]\!]$ is an interpretation function from sentences of a given language to propositions, such that:

If ϕ is atomic, then $[\![\phi]\!] = \langle X, Y \rangle$ for $X, Y \subseteq S$.

$[\![\neg\phi]\!] = \langle X, Y \rangle$ such that $[\![\phi]\!] = \langle Y, X \rangle$.

$[\![\phi \wedge \psi]\!] = \langle X, Y \rangle$ such that:

(i) $X = \{s \sqcup t : s \in [\![\phi]\!]^+ \text{ and } t \in [\![\psi]\!]^+\}$
(ii) $Y = [\![\phi]\!]^- \cup [\![\psi]\!]^-$

$[\![\phi \vee \psi]\!] = \langle X, Y \rangle$ such that:

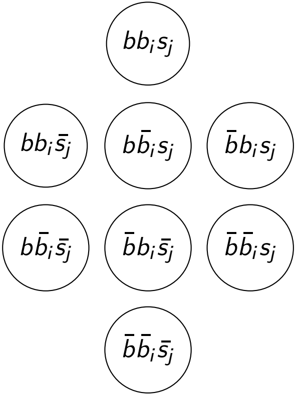
(i) $X = [\![\phi]\!]^+ \cup [\![\psi]\!]^+$
(ii) $Y = \{s \sqcup t : s \in [\![\phi]\!]^- \text{ and } t \in [\![\psi]\!]^-\}$

Appendix B

$$P^{+}(w) = \{s \in P^{+} : s \sqsubseteq w\}$$

$$P^{-}(w) = \{s \in P^{-} : s \sqsubseteq w\}$$

Appendix C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$\bar{b}b_is_j$
$b\bar{b}_i\bar{s}_j$
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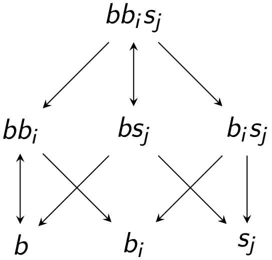
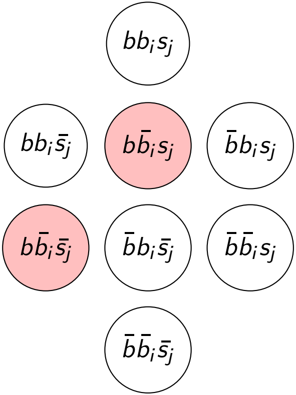
$\bar{b}\bar{b}_i\bar{s}_j$

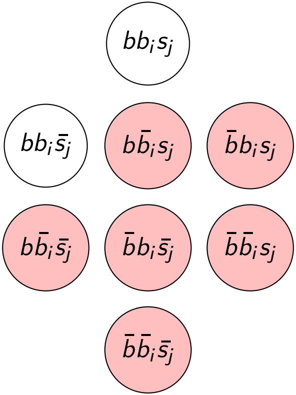

bb_is_j
bb_i
bs_j
b_is_j
b
b_i
s_j

Appendix C



$$
\begin{tikzcd}
 & bb_is_j \arrow[dl] \arrow[d, leftrightarrow] \arrow[dr] & \\
bb_i \arrow[d, leftrightarrow] \arrow[dr] & bs_j \arrow[dl] \arrow[dr] & b_is_j \arrow[dl] \arrow[d] \\
b & b_i & s_j
\end{tikzcd}
$$

Appendix C



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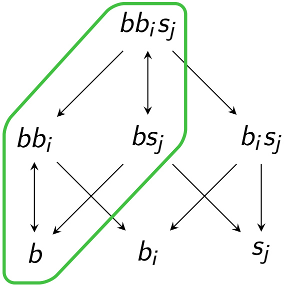
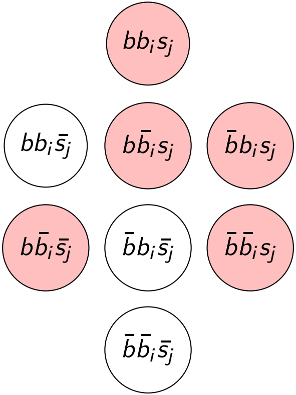
$\bar{b}b_is_j$
$b\bar{b}_i\bar{s}_j$
$\bar{b}b_i\bar{s}_j$
$\bar{b}\bar{b}_is_j$
$\bar{b}\bar{b}_i\bar{s}_j$



bb_is_j
bb_i
bs_j
b_is_j
b
b_i
s_j

Appendix C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$b\bar{b}_i\bar{s}_j$
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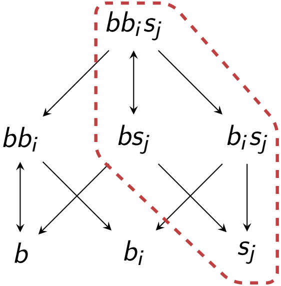
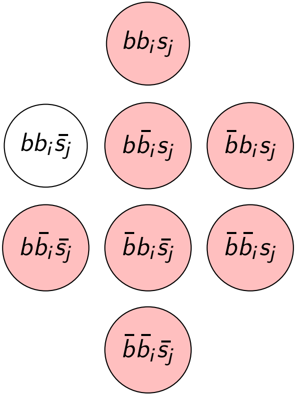

bb_is_j
bb_i
bs_j
b_is_j
b
b_i
s_j

Appendix C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$bb_i \bar{s}_j$
$b\bar{b}_i s_j$
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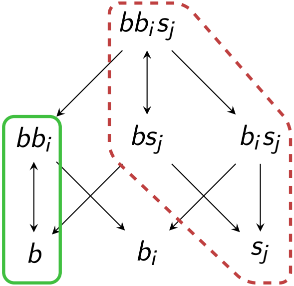
$b\bar{b}_i \bar{s}_j$
$\bar{b}b_i \bar{s}_j$
$\bar{b}\bar{b}_i s_j$
$\bar{b}\bar{b}_i \bar{s}_j$


$bb_i s_j$
bb_i
bs_j
$b_i s_j$
b
b_i
s_j